

James Sungarda

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Education

The University of Hong Kong

Hong Kong SAR

BACHELOR OF ENGINEERING (COMPUTER SCIENCE)

Sep. 2024 - Jun. 2028 (expected)

- Cumulative GPA: 3.42/4.30.

Skills

Programming Languages	Python, Javascript/Typescript, C++, HTML/CSS, SQL, Bash, Lua, TeX
Frameworks	Next.js, React, Django, Flask, Express, Matplotlib, Pandas, NumPy, PyTorch
Languages	English (Native), Mandarin (Fluent), Cantonese (Basic), Indonesian (Native)

Experience

Lalamove

Hong Kong SAR

INTERN, DRIVER OPERATIONS

Jun. 2026 - Aug. 2026

- Architected an automated document verification pipeline and engineered a dynamic supplemental submission field directly into the application, driving a **>45%** reduction in driver Vehicle Info Change (VIC) rejections. By addressing an **81%** systemic rejection rate, this enhancement reduced in-person escalations and reclaimed **400 minutes** of daily manual agent handling time.
- Audited the internal **AI Face Recognition** (FR) system, which accounted for the highest volume of agent tasks. Extracted large-scale FR match data from internal analytics tools into Excel and leveraged **NumPy** to model submission trends and identify algorithmic friction points. Defined key evaluation pipelines to benchmark human-AI matching rates, pitching system enhancements projected to yield a **30%** improvement in automated matching performance.
- Analysed onsite agent pipelines within the internal driver management tool to optimize FR handling. Proposed structural SOP improvements that decreased manual verification volume by **48%** (**~6,000 requests**) per day.

School of Computing and Data Science, HKU

Hong Kong SAR

RESEARCH ASSISTANT

Jun. 2025 - Dec. 2025

- Engineered and deployed a production-grade pedagogical tool supporting **300+ active users** across the entire HKU Department of Social Work and Social Administration; maintained **99.9%** uptime during peak curriculum integration periods.
- Architected and developed a high-traffic full-stack infrastructure using **Next.js, MongoDB, and TailwindCSS**, optimizing database queries to handle **500+ monthly sessions**.
- Developed scalable **RESTful API** endpoints with **Flask** to facilitate real-time LLM inferences, achieving an average latency of **<200ms** and successfully processing **6,000+** requests without service degradation.
- Published and presented a research paper on NLP applications as an educational tool as **first author**, accepted at **IEEE International Conference on Big Data 2025**, a top-tier academic conference in AI and Data Science with an acceptance rate of **18%**.
- Served as **Program Committee** (PC) Member for the Full Paper Track of **ACM International Conference on Web Search and Data Mining** (WSDM 2027), a flagship publication in AI and Data Science (ranked CORE A).

Publications

GenAI for Social Work Field Education: Client Simulation with Real-Time Feedback (<https://ieeexplore.ieee.org/document/11402062>)

IEEE BigData 2025

James Sungarda, HONGKAI LIU, ZILONG ZHOU, TIEN-HSUAN WU, JOHNSON C.S. CHEUNG, BEN KAO

8 Dec. 2025

- Explored novel methods of integrating cognitive models and learning pedagogies with Generative AI as an educational tool in the social work domain, focusing on **high-fidelity client simulations** and **real-time feedback mechanisms**.
- Trained **BERT-based language models** and **LLM few-shot learning** for multi-label classification, enhancing contextual awareness by using domain specific pre-training and achieving **~97%** accuracy rate.
- Organized and processed large scale **text annotation** tasks of over **13,000+** entries by collaborating with social work professionals to ensure high-quality labelled datasets for supervised learning.

Projects

Arang (<https://github.com/invalidcode232/arang>)

TYPESCRIPT, NEXT.JS, POSTGRESQL, AZURE

Oct. 2024

- Arang is a **spaced repetition** algorithm flashcards app which combines traditional digital flashcards functionality with LLM models using **Retrieval Augmented Generation** (RAG) to intelligently generate flashcards questions and answers based on images, text, and videos.
- **Full-stack web application** written in TypeScript, using the **Next.js framework with PostgreSQL** as backend database. The app connects to Azure's OpenAI service as the LLM provider.

Honors & Awards

Sep. 2025 **Third Prize**, iCANX Global Innovation Award 2025.

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Oct. 2024 **Second Runner Up**, HKU CSAxDSA Hackathon 2024.

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Sep. 2024 **Scholar**, HKU Undergraduate Entrance Scholarship for Outstanding Academic Talents.

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Jun. 2022 **Finalist**, National Olympiad in Informatics.

Indonesia